

# Seyyed Moein Kazemi

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[GitHub](#) | [Google Scholar](#) | [ResearchGate](#) | [LinkedIn](#)

## RESEARCH INTERESTS

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Deep Learning · Deep Reinforcement Learning · Medical AI · Efficient Deep Learning · Time Series Forecasting

## EDUCATION

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### Shahid Beheshti University

Sep 2024 – Current

*Master's Degree, Data Science*

*Tehran, Iran*

- **Thesis:** Designing a Dynamic Forecasting and Risk Management Framework in Stock Trading: A Probabilistic Deep Learning Approach (*Supervisor: Dr. Ehsan Bahrami Samani*)

### Iran University of Science and Technology

Sep 2019 – Nov 2023

*Bachelor's Degree, Computer Engineering*

*Tehran, Iran*

- **Thesis:** Storing Big Data on DNA: Design and Implementation of a Secure Pipeline Using Cryptographic Protocols (*Supervisor: Dr. Marzieh Malekimajd*)

## PUBLICATIONS & PREPRINTS

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1. **Kazemi, S. M.**, Azarpour, A., Riahi Madvar, M. CRNN-KAN: A Lightweight Hybrid Framework for Robust Phonocardiogram Classification. *Informatics in Medicine Unlocked*. Status: Under Review.
2. Azarpour, A, **Kazemi, S. M.** SW-KAN: Kolmogorov–Arnold Networks with Stieltjes–Wigert  $q$ -Orthogonal Polynomials. *Neurocomputing*. Status: Under Review.
3. Riahi Madvar, M., **Kazemi, S. M.**, Yousefi, S. Scalable unsupervised outlier detection in high-dimensional big data via collision-based subspace selection and deep reinforcement learning. *Applied Soft Computing*. Status: Under Review.
4. **Kazemi, S. M.**, Bahrami Samani, E. Probabilistic Joint Forecasting of Bivariate Time Series with LSTM via Dynamic Student- $t$  Likelihood. *Annals of Data Science*. Status: Under Review.
5. **Kazemi, S. M.**, Ghanbary, S., Riahi-Madvar, M. Discovering Actionable Marketing Insights: Rare Association Rule Mining for Customer Behavior Analysis. *2025 10th International Congress on Fuzzy and Intelligent Systems (CFIS)*.

## RESEARCH EXPERIENCE

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### Research Assistant

Apr 2025 – Sep 2026

*Supervisor: Dr. Mahboobeh Riahi-Madvar*

*Shahid Beheshti University, Tehran, Iran*

- **Lightweight Hybrid Deep Learning for Medical Time Series (CRNN-KAN)** (*Jan 2026 – Sep 2026*)
  - Proposed CRNN-KAN for robust phonocardiogram (PCG) classification, integrating a CNN-BiLSTM backbone with an EfficientKAN head (SiLU and sinusoidal bases).
  - Achieved competitive AUC scores across PhysioNet, CirCor, and PASCAL datasets, reducing trainable parameters to 1.31M (100x reduction vs. VGG16).
  - Validated physiological interpretability of learned acoustic representations via Score-CAM.
- **Scalable Outlier Detection in Big Data (mCRMR-GPPO)** (*Apr 2025 – Dec 2025*)
  - Developed a distributed unsupervised anomaly detection framework on Apache Spark, formulating an LSH-based mCRMR criterion for collision-based subspace selection.

- Designed a Gated Proximal Policy Optimization (GPPO) architecture to efficiently filter normal instances and dedicate DNN processing solely to complex anomalies.
- Demonstrated near-linear scalability and achieved  $\approx 99\%$  AUC on high-dimensional datasets.
- **Rare Association Rule Mining for Bank Marketing** *(Jul 2025 – Sep 2025)*
  - Designed an interpretable pipeline using dynamic minimum-support and MinHash LSH to efficiently extract actionable “flip rules” from high-impact rare patterns.
  - Identified precise decision boundaries that improved targeted marketing conversion rates from 11.7% to 64.7% (a 5.53x lift).

#### Research Assistant

Oct 2025 – Dec 2025

Supervisor: *Dr. Ehsan Bahrami Samani*

*Shahid Beheshti University, Tehran, Iran*

- **Probabilistic Joint Forecasting of Bivariate Time Series**
  - Engineered a Bivariate Probabilistic LSTM framework utilizing a dynamic Student- $t$  likelihood (optimized via NLL) to forecast returns, volatility, and cross-asset correlations for Gold and Brent Crude Oil.
  - Produced well-calibrated 95% predictive intervals, outperforming SOTA baselines (Transformer, TCN, N-BEATS) in accuracy and tail risk quantification.

#### Independent Researcher

Jan 2026 – Present

*Collaborative Research, Tehran, Iran*

- **Stieltjes-Wigert  $q$ -Orthogonal Polynomials in KANs (SW-KAN)** *(Jan 2026 – Sep 2026)*
  - Developed SW-KAN by replacing B-splines with Stieltjes-Wigert  $q$ -orthogonal polynomials, resolving domain mismatch via a smooth exponential-of-tanh mapping to  $(0, \infty)$ .
  - Leveraged  $\mathcal{O}(N)$  three-term recurrence, outperforming 18 baseline polynomial KAN variants on image classification and 2D function approximation tasks.

#### Undergraduate Research Assistant

Sep 2022 – Sep 2023

Supervisor: *Dr. Marzieh Malekimajd* *Iran University of Science and Technology, Tehran, Iran*

- **Secure Big Data Storage on DNA**
  - Developed a Python data-to-DNA simulation pipeline integrating file-to-binary conversion, Huffman coding, and sequence mapping.
  - Enhanced the DNACloud framework by implementing robust AES and OTP cryptographic layers, benchmarking payloads up to 10 million characters.

#### TEACHING & ACADEMIC EXPERIENCE

- **Teaching Assistant:** Machine Learning, Multivariate Analysis *(Instructor: Dr. Sakineh Dehghan)*
- **Teaching Assistant:** Machine Learning, Advanced Machine Learning, Pattern Recognition *(Instructor: Dr. Ahmad Ali Abin)*
- **Head TA & TA:** Advanced Data Mining (Head TA), Machine Learning *(Instructor: Dr. Mahboobeh Riahi-Madvar)*
- **Teaching Assistant:** Big Data *(Instructor: Dr. Hamed Malek)*
- **Teaching Assistant:** Deep Learning *(Instructor: Dr. Zeinab Hajimohammadi)*

#### SELECTED TECHNICAL PROJECTS

- **CXR-Report-Generator:** End-to-end encoder-decoder (ViT + GPT-2) via cross-attention for automated clinical report generation from Chest X-rays. [[GitHub](#)]

- **FracAtlas Bone Fracture CAD:** Multi-task diagnosis pipeline using DenseNet-121 (classification), ResNet34-UNet (segmentation), and Meta’s SAM. [[GitHub](#)]
- **DeepRC Immune Classifier:** Deep Multiple Instance Learning (1D CNN + LSTM + attention pooling) to predict disease status from immunosequencing data. [[GitHub](#)]
- **Multimodal Semantic Search:** Visual search engine leveraging OpenAI CLIP for semantic extraction and Qdrant vector database for K-NN retrieval. [[GitHub](#)]

## TECHNICAL SKILLS

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- **Programming Languages:** Python, C/C++, SQL, R, MATLAB, Bash.
- **AI & Deep Learning Libraries:** PyTorch (Geometric, Torchaudio, Torchvision), TensorFlow, Hugging Face, XGBoost, Scikit-Learn, OpenCV, CUDA.
- **Architectures & Paradigms:** Sequence Models (KAN, LSTM, TCN), Foundation Models (ViT, CLIP, SAM), Deep RL (PPO, DQN, Contextual Bandits), Probabilistic Modeling, LLM Agents & RAG.
- **Data Engineering & MLOps:** PySpark, Vector DBs (Qdrant, ChromaDB), Relational DBs (PostgreSQL), Docker, FastAPI, Linux.
- **Developer Tools:** Git, L<sup>A</sup>T<sub>E</sub>X, GNU Make.

## SELECTED COURSEWORK

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Machine Learning (19.75/20) · Deep Learning (19.5/20) · Advanced Data Mining (19.25/20) · Multi-Agent Systems [Focus: RL & LLMs] (19/20) · Artificial Neural Networks (18.5/20) · Differential Equations (19/20) · Multivariate Statistical Analysis (18.5/20) · Engineering Probability & Statistics (17/20) · Big Data Analytics (17.25/20)

## HONORS & AWARDS

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- **Top 1.5% Nationwide:** Among ~15,000 applicants in the Iranian University Entrance Exam for M.Sc., 2024.
- **Top 0.6% Nationwide:** Among >160,000 applicants in the Iranian University Entrance Exam for B.Sc., 2019.